

CommandIA

Version 1.0 (MVP Sprint)

Project Name: CommandIA

Mission: To eliminate social media sales friction for Algerian e-commerce businesses by deploying an AI assistant that handles inquiries, closes sales in DMs, and extracts structured, ready-to-ship orders.

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1. Product Overview & Value Proposition

The Problem

Algerian e-commerce relies heavily on Cash on Delivery (COD) via Meta platforms (Instagram/Facebook). Merchants spend excessive time manually answering repetitive questions (e.g., "Prix?", "Livraison?") and manually extracting unstructured delivery details from DMs, leading to high error rates and returned packages.

The Solution

CommandIA is a conversational engine that plugs into Meta APIs. It speaks the local dialects, answers product questions, guides the user to a purchase, and automatically extracts structured order data directly into a merchant dashboard.

2. Core Features (MVP Scope)

For the MVP, we are building three core components:

Component	Description
The Webhook Pipeline	Connecting Meta Graph API to our backend.
The AI Brain	The LLM engine trained on local dialects to converse and extract JSON data.
The Merchant Dashboard	A UI where merchants view orders and manage their product catalog.

3. Business Logic & AI Conversational Rules

3.1 Language & Identity Rules

Input Recognition: The AI must seamlessly process inputs in four languages/dialects:

- Algerian Darija (Arabic script)
- Franco-Arabic (Latin script with numerals, e.g., "bchhal", "3andkom")
- French
- English

Output Consistency: The AI should mirror the language of the user's initial message. If the user speaks Darija, reply in Darija. If they speak French, reply in French.

Tone: Polite, professional, and commercial. The primary directive is to close a confirmed order or answer a product inquiry quickly.

3.2 The "5 Sacred Fields" Data Extraction

The AI cannot mark an order as "Confirmed" until it successfully collects, validates, and stores these 5 specific fields:

#	Field	Validation Rule
1	Full Name	Must contain at least a first name.
2	Phone Number	Must be a valid Algerian number starting with 05, 06, or 07 (10 digits). If invalid, the bot must prompt the user to correct it.
3	Wilaya	Must match one of the 58 official Algerian Wilayas.
4	Delivery Type	The user must explicitly choose between: Home Delivery (à domicile / l'dar) or Desk/Office Delivery (Stop-desk / point relais / bureau) - only if the merchant has enabled this option.
5	Address (Commune/Street)	If Home Delivery: a detailed address (Commune, street, or clear location details). If Desk Delivery: the specific Commune or agency location.

3.3 Pricing & Negotiation Guardrails

- No Unauthorized Discounts:** The AI must never offer or accept a discount unless a specific rule is enabled in the merchant's dashboard.
- Haggle Defense:** If a user attempts to negotiate ("Nqassli chwiya?"), the AI politely declines and redirects to the purchase ("Les prix sont fixes, mais la qualité est garantie!").
- Dynamic Shipping Calculation:** The AI must add the correct shipping fee based on the selected Wilaya and Delivery Type (Home vs. Desk) to calculate the final Total Price.

4. System Architecture & Workflow State Machine

The AI must track the user's progress through a defined conversational state machine to prevent infinite loops.

State Definitions

State	Name	Behavior
State 0	Inquiry	User asks a question (price, size, color). AI identifies the product and answers.
State 1	Order Intent	User indicates they want to buy. AI begins the collection loop.
State 2	Data Collection	AI systematically asks for the 5 Sacred Fields (Name, Phone, Wilaya, Delivery Type, Address).
State 3	Confirmation	AI summarizes the order (Items, Delivery Type, Address, Total Price including shipping) and asks for final validation.

State	Name	Behavior
Escalation	Human Handoff	If the user is frustrated, asks for custom bulk pricing, or if the bot fails to extract valid data after 3 attempts, it pauses and flags the chat for human intervention.

5. Technical Requirements & Stack

Layer	Technology
Backend / Middleware	Python (FastAPI). Responsible for receiving Meta webhooks and managing the API pipeline.
AI Engine	OpenAI (gpt-4o-mini) or Anthropic (claude-3-5-haiku).
Data Parsing	Pydantic (Python) or Instructor to enforce strict JSON schemas for the 5 Sacred Fields.
Database	Supabase (PostgreSQL). Tables required: Merchants, Products, Orders, Conversations.
Frontend (Dashboard)	Next.js (React) + Tailwind CSS.

6. Success Metrics for MVP

To prove the viability of CommandIA during the ThirdUni program, we must achieve:

- **Successful Webhook Integration:** The system can reliably receive and reply to real Instagram DMs.
- **Extraction Accuracy:** The AI correctly formats the 5 Sacred Fields into JSON in 90%+ of test cases.
- **Merchant Onboarding:** The dashboard successfully connects to a test merchant's Meta account and displays incoming orders in real-time.